Hierarchical Shell Manifold Theory:
Graded Operator Reduction and the Physical-Selection Problem

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Abstract

Hierarchical Shell Manifold Theory (HSMT) is formulated as a graded operator framework for separating exact mathematical reduction from physical selection. The minimal structure is a Hilbert space, a family of mutually orthogonal grade projectors, and a reference operator. Standard functional-analytic tools then provide controlled self-adjoint realizations, exact Schur–Feshbach reduction, energy-dependent effective operators, exact memory kernels, and a scalar spectral inverse problem with an analytic semicircle benchmark. The HSMT-specific contribution of the present paper is an identifiability analysis of what these constructions do and do not determine physically. In particular, a grade label is not a physical dimension, a mathematical evolution parameter is not operational time, exact projection does not select the physical sector, mode-count scaling does not by itself determine physical dimension, and a positive-semidefinite or elliptic operator does not by itself define a Lorentzian causal cone. A monotone positive-coupling class is also shown unable to generate an interior minimum of the spectral bottom. The present framework does not yet select a unique physical realization and therefore makes no unique framework-level physical prediction. Specific conditional realizations are nevertheless testable once an empirical or microscopic selector is supplied. The paper is thus a foundations analysis of the physical-selection problem in a graded operator framework, not a completed derivation of observed spacetime.

Keywords: Foundations of physics; Graded operator framework; Effective operator reduction; Feshbach projection; Physical identifiability; Physical selection.

1 Introduction

A recurring problem in foundational model building is the gap between possessing a mathematical structure and deriving the physical meaning later assigned to it. A one-parameter unitary group supplies mathematical evolution, but not by itself an operational clock. A spectral exponent can characterize a response or mode count without thereby being the dimension of physical space. A reduced sector can possess exact effective dynamics even when the framework contains no principle selecting that sector as physically realized. These distinctions are especially important in programs that seek to treat spacetime structure as emergent rather than primitive.

HSMT was developed to study how much operator, spectral, and effective structure can be obtained before such physical identifications are made. Its mature core is

$$(\mathcal{H}, \{P_N\}, D), \tag{1}$$

where $\mathcal{H}$ is a complex Hilbert space, P_N are grade projectors, and D is a densely defined reference operator. The historical shell-manifold terminology is retained for continuity, but the mature foundational core is more abstract than a smooth-manifold model. A realization may attach geometric

objects $\mathcal{M}_N$ to the grades, but charts, smooth structure, metric signature, and physical dimension are not primitive assumptions of (1). Accordingly, "manifold" denotes a class of optional geometric realizations rather than a foundational smooth-manifold axiom.

The paper has two purposes. First, it presents the self-contained mathematical results supported by the present framework. Second, it treats physical identifiability as a separate problem: which physical interpretations are selected by the mathematics, and which remain underdetermined? The latter are not editorial caveats; they are the central foundations content of the paper because they distinguish exact reduction from physical selection. At the present stage HSMT does not make a unique framework-level physical prediction, since no unique physical realization has been selected.

1.1 Relation to existing mathematics

Several tools used below are standard. Bounded-perturbation self-adjointness is a direct application of Kato–Rellich perturbation theory [1, 2]. Projection and effective-operator methods belong to the Schur–Feshbach and projection-operator traditions [3, 4, 5, 6, 7, 8]. Stieltjes transforms and spectral measures are standard tools of self-adjoint spectral theory [2, 9]. Generalized measurements and informational completeness belong to the POVM formalism [10]. Scale-dependent spectral dimensions occur in several quantum-gravity programs [11, 12]. Green–Kubo formulas provide the standard transport criterion invoked in the diffusion discussion [13].

HSMT-specific content lies not in re-deriving these standard tools, but in their organization within a graded architecture, in the interpretive firewalls imposed on that architecture, in the class-specific monotonicity/no-selection results, and in the explicit separation between exact effective reduction and physical-sector selection.

Table 1: Standard mathematical ingredients versus the HSMT-specific role assigned to them in this paper.

Standard ingredient	**HSMT-specific contribution in the present paper**
Orthogonal Hilbert-space decomposition	grade architecture together with the explicit firewall that grade is not physical dimension
Kato–Rellich perturbation theory	controlled self-adjoint graded realizations without promoting the operator to a uniquely physical generator
Schur–Feshbach reduction	exact separation of mathematical reduction from the unresolved physical-sector selection problem
Stieltjes/spectral inversion	controlled response reconstruction while keeping spectral data distinct from physical geometric identification
Logarithmic scaling exponents	explicit distinction between a scale diagnostic and physical dimension
Unitary evolution and conserved observables	persistent-record structure without identifying the evolution parameter with operational time
Positive-semidefinite operator ordering	class-specific no-go result for monotone interior stabilization

2 Logical status and interpretive firewalls

We distinguish four logically different kinds of statement: theorem, no-go result, non-selection statement, and conditional physical bridge. A no-go result excludes a conclusion inside a declared hypothesis

class. A non-selection statement says only that the present premises do not determine a structure. A conditional bridge has the form "if an independently justified physical identification is supplied, then the following mathematical consequence may be interpreted physically."

The core firewalls are summarized in Table 2.

Table 2: Interpretive firewalls.

Object	**Mathematical role**	**Not derived automatically**
Grade N	hierarchy label	physical dimension
Evolution parameter τ	parameter of a one-parameter evolution	operational physical time
Generator H or D	spectral/evolution operator	physical clock or Lorentzian metric
Mode or spectral scaling	diagnostic of specified data	observed spatial dimension
Projected sector $P\mathcal{H}$	reduced mathematical sector	uniquely realized physical world
Propagation/principal form	mathematical characteristic information	Lorentzian causal cone or physical light speed
Conserved observable	persistent record quantity	arrow of time or irreversible measurement

The guiding rule is simple: a mathematical object is not a derived physical observable merely because it can be renamed after one.

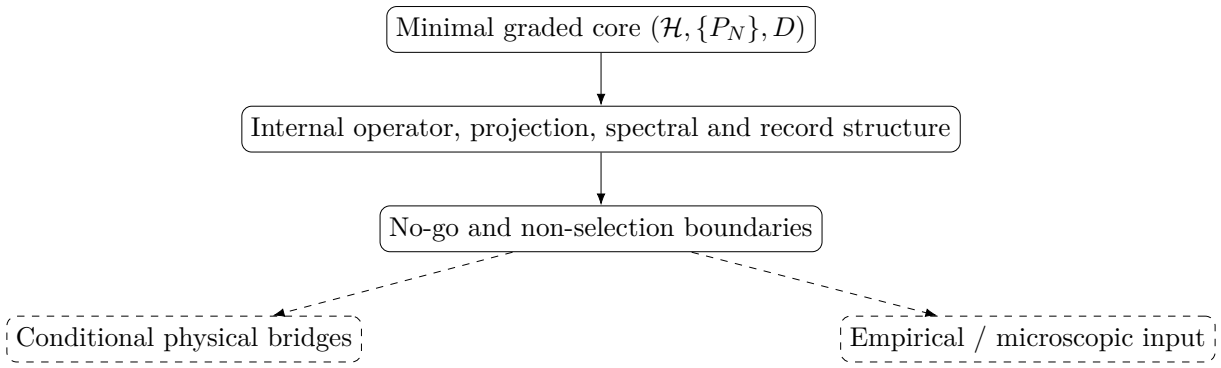


Figure 1: Logical architecture. Solid arrows denote internal mathematical dependence. Dashed arrows mark the presently open transition from internal structure to physical interpretation.

3 Minimal graded architecture

Let $\mathcal{H}$ be a complex separable Hilbert space and let $\{P_N\}_{N\in\mathcal{I}}$ be a countable family of mutually orthogonal projections satisfying

$$P_N P_M = \delta_{NM} P_N, \qquad \sum_{N\in\mathcal{I}} P_N = I \tag{2}$$

in the strong operator topology.

Set

$$\mathcal{H}_N = P_N \mathcal{H}, \tag{3}$$

so that

$$\mathcal{H} = \bigoplus_{N \in \mathcal{I}} \mathcal{H}_N. \tag{4}$$

The label N is the grade. Nothing in this definition assigns a physical dimension to N.

Let D be a densely defined reference operator. For an operator A whose blocks are well defined, write

$$A_{NM} = P_N A P_M. \tag{5}$$

A nearest-neighbor restriction, such as

$$A_{NM} = 0 \qquad \text{for } |N - M| > 1, \tag{6}$$

is an additional model assumption rather than a consequence of grading.

For the class of closed evolutions considered here, norm preservation is the only dynamical axiom. In a self-adjoint realization,

$$U(\tau) = e^{-\mathrm{i}H\tau}, \qquad \|U(\tau)\psi\| = \|\psi\|. \tag{7}$$

The symbol τ is deliberately not called physical time.

4 Self-adjoint graded realizations

The following standard theorem provides a clean operator-control class.

Theorem 4.1 (Bounded intergrade perturbation)**.** *Let D_0 be self-adjoint on $\mathrm{Dom}(D_0) \subset \mathcal{H}$ and let T be bounded and self-adjoint on $\mathcal{H}$. Then*

$$D = D_0 + T \tag{8}$$

is self-adjoint on $\mathrm{Dom}(D_0)$.

Proof. A bounded self-adjoint T is D_0-bounded with relative bound zero. Kato–Rellich therefore applies [1, 2]. □

If D_0 is grade diagonal and T intergrade, this theorem supplies a controlled graded realization. It does not select a unique physical generator.

5 Projection, effective operators, and memory

Because unbounded block-operator decompositions require domain care, we state the exact reduction in a deliberately controlled form.

Theorem 5.1 (Exact projected resolvent in a block-controlled realization)**.** *Let*

$$\mathcal{H} = \mathcal{H}_P \oplus \mathcal{H}_Q. \tag{9}$$

Let A and C be self-adjoint on $\mathcal{H}_P$ and $\mathcal{H}_Q$, respectively, and let

$$B : \mathcal{H}_Q \to \mathcal{H}_P \tag{10}$$

be bounded.

Define

$$H = \begin{pmatrix} A & B \\ B^\dagger & C \end{pmatrix}, \qquad \mathrm{Dom}(H) = \mathrm{Dom}(A) \oplus \mathrm{Dom}(C). \tag{11}$$

Then H is self-adjoint.
Let $z \in \rho(C)$ and define

$$F_P(z) = z - A - B(z - C)^{-1}B^\dagger. \tag{12}$$

Then $z \in \rho(H)$ if and only if $F_P(z)$ is invertible on $\mathcal{H}_P$, and in that case

$$P(z - H)^{-1}P = F_P(z)^{-1}. \tag{13}$$

Proof. Self-adjointness follows by treating the bounded off-diagonal block as a bounded perturbation of $A \oplus C$. Equation (13) is then the Schur complement of $z - H$; see [3, 4, 5]. □

The exact self-energy is

$$\Sigma_P(z) = B(z - C)^{-1}B^\dagger, \tag{14}$$

so

$$H_{\mathrm{eff}}(z) = A + \Sigma_P(z). \tag{15}$$

For z off $\mathrm{spec}(C)$,

$$\|\Sigma_P(z)\| \le \frac{\|B\|^2}{\mathrm{dist}(z, \mathrm{spec}(C))}. \tag{16}$$

For the corresponding unitary evolution, the state satisfies

$$\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}\tau}\Psi(\tau) = H\Psi(\tau), \qquad \Psi = (\psi_P, \psi_Q), \tag{17}$$

on the appropriate domain.

Solving the Q-sector equation gives

$$\psi_Q(\tau) = e^{-\mathrm{i}C(\tau-\tau_0)}\psi_Q(\tau_0) \tag{18}$$

$$- \mathrm{i}\int_{\tau_0}^{\tau} e^{-\mathrm{i}C(\tau-s)}B^\dagger\psi_P(s)\,\mathrm{d}s, \tag{19}$$

and substitution yields an exact P-sector memory kernel

$$K(\Delta\tau) = Be^{-\mathrm{i}C\Delta\tau}B^\dagger. \tag{20}$$

This is the same structural origin of memory emphasized in projection-operator approaches [6, 7, 8].

Boundary Result 5.2 (Projection is not physical selection)**.** *The reduction assumes P. Equations* (12)–(20) *determine the exact reduced dynamics of a chosen sector but provide no principle selecting that sector as physically realized.*

6 Spectral inverse structure and an analytic benchmark

For the scalar realization used in the benchmark program,

$$G(z) = \frac{1}{z - \omega_0 - \Sigma(z)}, \tag{21}$$

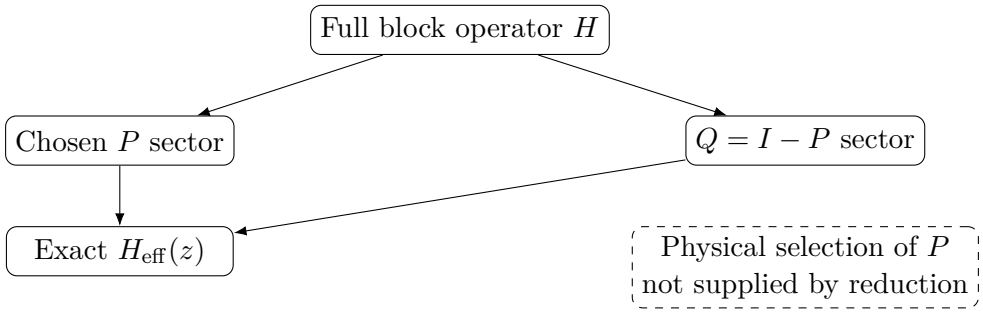


Figure 2: Exact reduction and physical selection are different logical operations.

so wherever $G(z) \neq 0$,

$$\Sigma(z) = z - \omega_0 - G(z)^{-1}. \tag{22}$$

Let μ be a positive spectral measure and

$$m(z) = \int \frac{\mathrm{d}\mu(M)}{z - M}, \qquad \Sigma(z) = g^2 m(z). \tag{23}$$

At regular points of an absolutely continuous part, the boundary-value inversion is

$$\rho(E) = -\frac{1}{\pi g^2} \lim_{\eta \to 0^+} \operatorname{Im} \Sigma(E + \mathrm{i}\eta), \tag{24}$$

a standard Stieltjes inversion statement [2, 9].

For the normalized semicircle

$$\rho_{\mathrm{sc}}(M) = \frac{\sqrt{4 - M^2}}{2\pi} \, \mathbf{1}_{[-2,2]}(M), \tag{25}$$

the transform is

$$m_{\mathrm{sc}}(z) = \frac{z - \sqrt{z^2 - 4}}{2}, \tag{26}$$

with the branch

$$\sqrt{z^2 - 4} \sim z \tag{27}$$

at infinity.

Its Fourier transform is

$$\int_{-2}^{2} \rho_{\mathrm{sc}}(M) e^{-\mathrm{i}M\tau} \, \mathrm{d}M = \frac{J_1(2\tau)}{\tau}, \tag{28}$$

so the memory envelope decays as

$$\tau^{-3/2} \tag{29}$$

up to oscillation.

The analytic relations above provide a self-contained benchmark for the inverse map and memory asymptotics. Numerical realization tests, including finite-resolution reconstruction, noise propagation, and pole searches, are reported separately in the benchmark companion, where their reproducibility requirements can be treated explicitly.

7 Scale diagnostics and dimensional firewalls

For a specified positive scale-dependent quantity $Q_N(\ell)$, define

$$d_N(\ell) = \frac{\mathrm{d}\ln Q_N(\ell)}{\mathrm{d}\ln \ell}. \tag{30}$$

If

$$Q_N(\ell) = C\ell^d \tag{31}$$

on an interval, then

$$d_N(\ell) = d \tag{32}$$

there.

This is a statement about logarithmic scaling. It becomes a statement about physical dimension only after the observable and scale variable have been given an independently justified geometric meaning.

HSMT therefore distinguishes possible notions such as

$$d_{\mathrm{geom}}, \qquad d_{\mathrm{res}}, \qquad d_{\mathrm{spec}}, \qquad d_{\mathrm{dyn}}, \tag{33}$$

and no general equality among them is assumed.

This caution is consistent with the broader quantum-gravity literature, where spectral dimension is a particular diagnostic and may run with scale [11, 12].

Proposition 7.1 (Mode-count scaling is not physical dimension)**.** *If*

$$Q(M) \sim CM^k, \qquad C > 0, \tag{34}$$

then

$$\frac{\mathrm{d}\ln Q}{\mathrm{d}\ln M} \longrightarrow k. \tag{35}$$

This determines a count exponent with respect to M. It does not establish physical dimension k unless M and Q are independently given the required operational geometric interpretation.

In particular, neither a monotone sequence of physical dimensions across grades nor a fractional interpolation between neighboring grades is derived by (30). Such behavior may be imposed in a model, but it is not selected by the core.

8 Closed evolution and persistent records

Let H be self-adjoint and

$$U(\tau) = e^{-\mathrm{i}H\tau}. \tag{36}$$

If a bounded observable R commutes with H—or, for unbounded observables, the appropriate strong-commutation conditions hold—then

$$U(\tau)^\dagger R U(\tau) = R, \tag{37}$$

and hence

$$\mathrm{Tr}[\rho(\tau)R] \tag{38}$$

is constant under unitary evolution.

This provides a clean mathematical notion of a persistent record quantity. It does not by itself provide an operational clock, temporal metric, irreversible measurement, or thermodynamic arrow. Generalized measurement theory supplies the standard POVM language for discussing measurements [10]. No HSMT-specific POVM theorem is used in the present paper: the previously recorded HSMT construction is retained only in the archival supplement until its exact operator formula and proof are recovered in self-contained form.

9 A monotone spectral no-go result

The strongest self-contained class-specific no-go result concerns monotonicity of the spectral bottom and can be stated without differentiability assumptions.

No-Go Result 9.1 (Monotone positive-coupling class)**.** *Let*

$$H(q) = H_* + \sum_e f_e(q_e) O_e, \tag{39}$$

where each

$$O_e \geq 0 \tag{40}$$

and each f_e is nondecreasing.

If

$$q'_e \geq q_e \tag{41}$$

componentwise, then

$$H(q') - H(q) = \sum_e [f_e(q'_e) - f_e(q_e)] O_e \geq 0. \tag{42}$$

Therefore

$$\inf \operatorname{spec} H(q') \geq \inf \operatorname{spec} H(q). \tag{43}$$

Consequently this class cannot, by itself, generate an interior minimum of the spectral bottom along a coordinate on which all couplings enter monotonically with fixed positive-semidefinite operators.

Proof. The operator inequality follows term by term. Monotonicity of the bottom of the spectrum follows from the variational principle. □

Where a differentiable nondegenerate ground-state branch exists, Hellmann–Feynman gives the equivalent local relation

$$\partial_{q_e} E_0 = f'_e(q_e) \langle O_e \rangle_0 \geq 0. \tag{44}$$

This is a class-specific spectral no-go theorem. In a model where the spectral bottom is interpreted as the relevant vacuum energy, it obstructs the desired interior stabilization. It does not exclude stabilization after new sign structure, nonmonotone dependence, additional dynamical variables, or other microscopic input is introduced.

10 Transport no-go and requirements for causal interpretation

Locality, dephasing, or clustering alone do not imply diffusion. A local unitary system can be ballistic, diffusive, anomalous, or localized depending on conserved quantities and current correlations. A genuine diffusive limit requires, among other conditions, the absence of a nonzero ballistic/Drude contribution

and suitable integrability of connected current correlations, as in Green–Kubo theory [13]. This is a mathematical no-go against inferring diffusion from locality and dephasing alone.

10.1 Requirements for causal interpretation

The causal discussion contains one mathematical non-implication and several identifiability requirements.

A positive-semidefinite or elliptic principal operator does not by itself supply a Lorentzian signature or causal cone: embedding that operator in a hyperbolic wave equation is additional dynamical structure.

Beyond that mathematical point, any physical causal interpretation requires at least the following:

(i) a propagation observable must be invariantly meaningful rather than a bare numerical speed;

(ii) the grade-to-characteristic map must be operationally discriminating if it is to identify a grade regime;

(iii) regime identification need not determine an absolute grade origin;

(iv) a postulated grade-dependent speed does not become a derived law unless its grade dependence, scale calibration, and observable consequences are independently obtained;

(v) a physical signal speed requires a relative normalization of temporal and spatial principal terms;

(vi) recovery of a causal cone and recovery of physical dimension are logically distinct identification problems.

These are requirements for interpretation, not all theorem-level no-go statements. Their verbatim historical formulations are retained in the archival supplement.

11 The physical-sector boundary

The exact effective theory of Eq. (13) is conditional on a chosen projector.

This motivates the central physical-selection problem:

Open Problem 11.1 (Physical sector). What empirical, microscopic, or independently justified principle identifies the physically realized grade/projector structure rather than merely analyzing a chosen mathematical decomposition?

The current framework does not answer this question.

Multi-contact response can constrain and falsify a proposed decomposition, but reconstruction of consequences from a chosen P is logically different from derivation of P.

This distinction propagates to the wider interpretive problem. The current premises do not derive, in a physically operational sense,

- a unique spacetime realization,
- physical dimension,
- Lorentzian causal structure,
- or common metric structure across independently emergent sectors.

12 Conditional bridges and falsifiability

The remaining physical program is conditional rather than hidden.

Examples are:

- if a projector is independently identified as physical, the Feshbach construction gives its exact reduced operator under the stated hypotheses;
- if a spectral diagnostic is operationally linked to geometry, its scale dependence may acquire geometric meaning;
- if the evolution parameter is linked to reproducible clocks and records with the required normalization, a physical-time interpretation may be investigated;
- if a characteristic structure is supplemented by an independently justified signal-propagation principle and temporal-spatial normalization, a causal interpretation may be investigated.

At present there is no unique framework-level physical prediction because the abstract framework does not yet select a unique physical realization.

Falsifiability must therefore be assigned to the correct logical layer:

(1) **Framework level:** no single presently derived observable uniquely tests the entire abstract HSMT class.

(2) **Theorem level:** class-specific mathematical claims are falsifiable by counterexample or failure of their stated hypotheses/conclusions.

(3) **Realization level:** a specified conditional realization is empirically testable and may be falsified without eliminating every realization admitted by the abstract framework.

(4) **Physical-bridge level:** any proposed selector becomes empirically vulnerable once it yields sufficiently specific operational predictions.

Thus specific conditional realizations are testable even though the abstract framework does not yet select a unique physical realization.

13 Discussion

The central HSMT-specific lesson is an identifiability result rather than a claim that the standard operator tools themselves are new: effective mathematical reduction and physical selection are distinct achievements. A graded system can possess exact reduced dynamics, memory kernels, inverse spectral maps, scale diagnostics, and conserved records while still failing to determine which sector or diagnostic should be identified with the observed world.

This perspective makes the negative results constructive. The monotone-coupling theorem identifies a broad mechanism that cannot stabilize the desired interior configuration. The mode-count firewall prevents a combinatorial exponent from being promoted into physical dimension without an operational bridge. The projection boundary prevents exact effective dynamics from being misread as physical-sector selection. The characteristic/causal sequence prevents elliptic or spatial principal structure from being promoted directly into Lorentzian causality.

Within the classes and branches analyzed here, no further internally licensed route to the missing physical selector has been identified. This is a methodological boundary rather than an absolute completeness claim. New empirical information, a new microscopic law, or an independently motivated external mathematical principle could reopen the derivation. Reparameterization, relabeling, or inserting the desired interpretation by hand would not.

14 Conclusion

HSMT provides a compact graded operator architecture supporting rigorous effective reduction and a controlled spectral inverse program. Its strongest present contribution is two-sided: it shows what follows from the graded premises and, with equal care, what does not. Grade is not dimension by definition; projection is not selection; mathematical evolution is not operational time; count scaling is not automatically spatial dimension; and elliptic or propagation structure is not automatically Lorentzian causality.

The framework therefore remains physically incomplete in a precise sense. A physical-selection criterion linking the mathematical architecture to observed spacetime has not been derived, and consequently HSMT presently makes no unique framework-level physical prediction. That gap should be treated as an explicit scientific problem rather than hidden inside notation. Specific conditional realizations can nevertheless be tested, and the benchmark program developed alongside the framework is reported separately. The full internal checkpoint history and source-fidelity record are retained in an archival supplement.

A Result-status summary

Result	**Status**	**Interpretive limit**
Bounded intergrade perturbation	theorem	no unique physical generator
Schur–Feshbach reduction	exact under block hypotheses	chosen P, not physical selection
Memory kernel	exact under same realization	no microscopic arrow of time
Scalar $G \to \Sigma$ inversion	exact in scalar model	no physical measure selection
Semicircle transform and memory law	exact analytic benchmark	inverse-map check, not physical measure selection
Logarithmic scale diagnostic	definition	not physical dimension without a bridge
Monotone positive-coupling result	no-go in declared class	not a universal no-vacuum theorem
Mode-count exponent	mathematical scaling	not physical dimension
Physical projector	open	requires empirical/microscopic input
Causal/dimensional bridge	open	characteristic and dimension recovery remain distinct

Statements and Declarations

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Data Availability. No new datasets were generated or analyzed in this study.

Use of generative AI tools. During manuscript preparation, the author used OpenAI ChatGPT and xAI Grok as analytical and editorial assistants for manuscript organization, drafting, mathematical exposition, reference checking, and critical review. All scientific claims, equations, citations, revisions, and the final manuscript were reviewed and approved by the author, who assumes full responsibility for the content.

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